

Practical Control Systems Interview Questions

These questions focus on practical control-system reasoning. In safety-critical applications, validate designs against the relevant standards, plant limits, and test procedures.

1. What is a feedback control system?

It measures an output, compares it with a reference, and uses the error $e(t) = r(t) - y(t)$ to adjust the input. Negative feedback can reject disturbances and reduce sensitivity to plant variation.

2. What is the difference between open-loop and closed-loop control?

Open-loop control commands an input without measuring the result. Closed-loop control uses output feedback, improving accuracy and disturbance rejection but adding stability, sensor, and tuning concerns.

3. What do proportional, integral, and derivative actions do?

For $u(t) = K_p e(t) + K_i \int e(t) dt + K_d de(t)/dt$, proportional action responds to present error, integral removes steady-state error, and derivative anticipates error changes. Each term can also worsen noise, overshoot, or stability if poorly tuned.

4. Why can integral action cause overshoot?

The integrator retains accumulated error while the actuator is saturated or the plant is slow. That stored command continues driving the plant after it crosses setpoint, a phenomenon called integral windup.

5. How do you prevent integral windup?

Limit the integrator, stop or back-calculate integration during actuator saturation, and initialize/reset it appropriately. The anti-windup method must use the actual actuator limits.

6. Why is derivative action often filtered?

An ideal derivative amplifies high-frequency measurement noise. A practical form is $K_d s / (1 + s/\omega_f)$, which limits high-frequency gain while preserving useful phase lead near crossover.

7. Why take derivative on measurement rather than on error?

Derivative on error produces a large “derivative kick” when the setpoint steps. Derivative on measurement avoids that kick while still responding to rapid output motion.

8. What is steady-state error?

It is $\lim_{t \rightarrow \infty} e(t)$ when the closed loop is stable and the limit exists. It depends on reference type, loop type, disturbances, and DC loop gain.

9. What is the final value theorem, and when is it valid?

$\lim_{t \rightarrow \infty} x(t) = \lim_{s \rightarrow 0} sX(s)$ when all poles of $sX(s)$ are in the open left half-plane, with at most a simple pole at the origin. Do not apply it to unstable or sustained-oscillation responses.

10. What does a pole represent physically?

Poles describe a system's natural modes. In continuous time, a pole with negative real part decays, a positive real part grows, and an imaginary-axis pole is marginal and needs special consideration.

11. What do zeros do to a response?

Zeros shape transient response and frequency response. A right-half-plane zero is nonminimum phase: it adds phase lag and can force an initial response opposite to the final direction.

12. What is a transfer function?

For zero initial conditions, $G(s) = Y(s)/U(s)$. It compactly represents a linear time-invariant input-output relationship but does not by itself expose all internal states or nonlinear behavior.

13. How is a closed-loop transfer function derived for unity negative feedback?

With plant $G(s)$ and controller $C(s)$, $T(s) = Y(s)/R(s) = C(s)G(s)/(1 + C(s)G(s))$. The denominator $1 + C(s)G(s)$ determines closed-loop characteristic roots.

14. What is loop gain?

The loop transfer function is $L(s) = C(s)G(s)H(s)$, including sensor feedback H . Its magnitude and phase determine stability margins, bandwidth, sensitivity, and disturbance rejection.

15. What do gain margin and phase margin mean?

Gain margin is the gain increase tolerated at the phase-crossover frequency before instability. Phase margin is additional phase lag tolerated at gain crossover; both are frequency-domain robustness indicators, not complete guarantees.

16. Why can a system with positive phase margin still perform poorly?

Margins may not cover nonlinearities, delays, multiple crossovers, uncertain dynamics, saturation, sampling effects, or performance requirements. Inspect time response and robustness over expected operating conditions.

17. What does a Bode plot show?

It plots magnitude and phase versus logarithmic frequency. It helps identify bandwidth, resonances, crossover, roll-off, phase lag, and approximate stability margins.

18. What is bandwidth in a closed-loop system?

It is commonly the frequency range over which the closed-loop response remains near its low-frequency gain, often to the -3 dB point. Higher bandwidth generally improves tracking but increases noise sensitivity and actuator demand.

19. What is the sensitivity function?

$S(s) = 1/(1 + L(s))$. It describes sensitivity to plant changes and input disturbances entering near the output; small $|S|$ at low frequency usually improves regulation.

20. What is complementary sensitivity?

$T(s) = L(s)/(1 + L(s)) = 1 - S(s)$. It describes reference tracking and measurement-noise transmission in common feedback configurations; high bandwidth can pass more sensor noise.

21. What is the “waterbed effect” in feedback design?

Because of fundamental sensitivity constraints, reducing sensitivity strongly in one frequency range often increases it elsewhere. Robust design balances low-frequency disturbance rejection against resonant peaking and uncertainty.

22. What is root locus used for?

It shows how closed-loop poles move as a scalar loop gain changes. Use it to assess stability ranges, damping, and approximate gain choices; verify the resulting transient and frequency response.

23. What is damping ratio?

For a standard second-order denominator $s^2 + 2\zeta\omega_n s + \omega_n^2$, ζ controls oscillation. For a step input with no finite zeros and $0 < \zeta < 1$, percent overshoot is approximately $100e^{-\pi\zeta/\sqrt{1-\zeta^2}}$.

24. What is natural frequency?

In the same second-order model, ω_n sets the basic time scale. It is not automatically the closed-loop bandwidth when zeros, higher-order modes, delays, or nonlinear limits are important.

25. Why is a pure time delay difficult to control?

A delay contributes phase lag $-\omega\tau$ without reducing gain. As frequency rises, it erodes phase margin and limits attainable bandwidth.

26. How would you identify an unknown plant safely?

Start with a low-risk model from physics or data, apply bounded test signals within operating limits, log inputs and outputs, and validate on separate data. Include delays, saturation, and operating-point dependence.

27. Why use a step test, and what are its limitations?

A step is simple and reveals gain, delay, time constants, and overshoot. It may violate constraints, poorly excite some dynamics, and give misleading results for nonlinear or drifting plants.

28. What is system order?

It is the number of independent dynamic states, often the highest denominator power after cancellations in a minimal transfer function. Higher order can introduce additional lags, resonances, and unmodeled dynamics.

29. What is controllability?

For $\dot{x} = Ax + Bu$, a state-space system is controllable if inputs can move the state between required states. The controllability matrix $[B \ AB \ \dots \ A^{n-1}B]$ must have full rank for an n -state LTI system.

30. What is observability?

For $y = Cx$, observability means internal states can be inferred from known inputs and outputs. The matrix $[C^T \ (CA)^T \ \dots \ (CA^{n-1})^T]^T$ must have full rank for an n -state LTI system.

31. Why use a state observer?

Some useful states are not measured directly or are too noisy to measure. An observer combines the model, input, and measurements to estimate them; its poles must balance speed, noise, and model error.

32. What is a Kalman filter in control applications?

It is an observer that estimates states using a stochastic model and measurement noise covariance. Its quality depends on realistic process-noise and sensor-noise models, not just mathematical implementation.

33. What is feedforward control?

It calculates an input from a known reference or measurable disturbance before an error appears.

It improves response when the model is accurate, but feedback is still needed for uncertainty and unmeasured disturbances.

34. How does cascade control help?

An inner fast loop regulates an intermediate variable, while an outer slower loop commands its setpoint. The inner loop should be tuned and validated first and have significantly greater bandwidth than the outer loop.

35. When is ratio control useful?

It maintains a proportional relationship between streams, such as fuel and air. It requires reliable flow measurements, appropriate bias/trim, and safety limits independent of the ratio command.

36. What is a deadband, and why is it used?

It is a range around setpoint where no corrective action is taken. It reduces actuator chatter and sensitivity to noise, but it permits a bounded steady-state error.

37. What is hysteresis in on-off control?

It uses different turn-on and turn-off thresholds. For example, a heater can turn on below 19°C and off above 21°C, preventing rapid switching near one threshold.

38. What is actuator saturation?

It occurs when an actuator reaches a physical amplitude, rate, or power limit. The controller must recognize these limits because saturation changes the effective loop and can cause windup or limit cycles.

39. What is rate limiting?

It limits how quickly an actuator command may change. It protects hardware and reflects real actuator dynamics, but adds lag that must be included in stability and tracking analysis.

40. Why can a valve with stiction cause oscillation?

Static friction makes small controller changes ineffective until force exceeds breakaway friction; then the valve jumps. The resulting nonlinear limit cycle may look like poor PID tuning, so diagnose mechanical behavior first.

41. What is a limit cycle?

It is a self-sustained bounded oscillation, often produced by nonlinearities such as relay action, quantization, backlash, saturation, or friction. Changing linear gains alone may not eliminate its cause.

42. What is sampling frequency, and how do you choose it?

It is the rate at which a digital controller samples signals. Choose it substantially faster than the desired closed-loop bandwidth and relevant plant dynamics, while accounting for computational delay, noise, and aliasing.

43. What is zero-order hold?

It holds a digital command constant between samples. This introduces effective phase lag and magnitude effects, which become significant as loop bandwidth approaches the sampling frequency.

44. Why add an anti-alias filter before an ADC?

Content above half the sample frequency folds into lower frequencies and cannot be removed digitally afterward. An analog low-pass filter attenuates that content before sampling.

45. How does quantization affect a control loop?

Finite sensor and actuator resolution creates small nonlinear errors, possible chatter, and limit cycles. Use adequate resolution, scaling, filtering, deadbands, or dither only after assessing noise and accuracy needs.

46. What is bumpless transfer?

It is switching between manual/automatic or between controllers without a command jump. Align the controller integral/state with the current actuator output before enabling the new mode.

47. Why should controller output be tracked in manual mode?

Tracking keeps the internal integral or bias consistent with the operator's command. Without it, returning to automatic can produce a sudden actuator step.

48. How should a PID controller be tuned in practice?

Establish safe limits, characterize the plant near its operating point, start conservatively, tune proportional response, add integral for required offset removal, then derivative/filtering only if justified. Validate disturbances, setpoint changes, noise, saturation, and fault cases.

49. Why is Ziegler–Nichols tuning not always appropriate?

Its aggressive settings can produce overshoot and poor robustness, and the test may be unsafe for real equipment. Use it only as a bounded starting point when its assumptions and risk are acceptable.

50. What is gain scheduling?

It changes controller parameters based on a measured operating condition such as speed, load, or temperature. It is useful for nonlinear plants, but transitions must be smooth and every schedule region validated.

51. How do you validate a control system before deployment?

Review requirements and limits, simulate nominal and worst cases, test hardware-in-the-loop where appropriate, exercise sensor/actuator faults and mode transitions, and commission gradually with independent safety protections active.

52. What is the first response to an unstable loop in a live plant?

Put the process into a documented safe state, often by switching to a safe manual output or shut-down procedure. Do not continue experimental tuning while oscillations could exceed equipment, process, or personnel limits.